

Experimental study on the anchoring mechanism and performance of BFRP bolt-anchorage system in supporting clay slopes

Jibin Chen^{a,b,c,d,*}, Xiaobo Lv^e, Jingwen Kang^{b,c}, Guanglei Liu^c, Yongxiang Zhou^d

^a Chengdu Technological University Research Institute, Chengdu Technological University, Yibin, 644000, China

^b China Southwest Geotechnical Investigation & Design Institute Co, Chengdu, 610052, China

^c Zhongyan Technology Co., Ltd, Beijing, 101100, China

^d Faculty of Architecture, Civil and Transportation Engineering, Beijing University of Technology, Beijing, 101124, China

^e School of Mechanical and Electrical Engineering, Yibin University, Yibin, 644000, China

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ABSTRACT

The experimental studies were performed to examine the failure mechanism and the capacity of BFRP bolt-anchorage system under laboratory and field conditions in supporting clay slopes in Sichuan Basin, China. The results indicate that BFRP anchor bolts, designed based on the principle of equal strength replacement between bolt tensile strength and the bonding strength of the first interface, can meet the safety standards required for slope engineering. During the stable phase of the slope, the mechanical behavior and deformation characteristics of BFRP anchor bolts are comparable to those of steel anchor bolts, with the axial force of BFRP bolts being 1/3 to 1/4 lower than the designed value. When the slope enters the accelerated creep stage, the axial force of steel anchor bolts exceeds the designed value by 40 %, while the axial force of BFRP bolts remains at only 2/3 of that of steel bolts. The failure mechanisms of the BFRP bolt-anchorage system primarily involve shear failure at the bolt-mortar interface and pullout failure of the bolt body, which are attributed to the cumulative damage of the polymer material. Based on the experimental findings, it is recommended that the minimum tensile safety factor for BFRP bars used in temporary slope support should be set at 1.26. This study enhances the understanding of BFRP anchorage systems in clay soil environments and provides valuable insights for the design and construction of infrastructure projects in similar geological conditions.

1. Introduction

Basalt Fiber Reinforcement Polymer (BFRP) bars are characterized by high strength, high corrosion resistance, and excellent environmental protection performance. It can replace steel bars as anchor bolts to promote the realization of “Net Zero emission” in geotechnical anchoring engineering. However, as a new type of construction material in civil engineering, the anchoring performance of BFRP anchor bolts has not yet been clarified, which hinders its application in geotechnical engineering to some extent. Therefore, investigating the design parameters and anchoring performance of BFRP anchor bolts in slope reinforcement projects is essential for establishing new material design and evaluation methods, which is conducive to mitigating geological disasters induced in geotechnical engineering and achieving sustainable development of green buildings.

In the field of geotechnical engineering, BFRP is commonly used for

soil reinforcement [1,2], concrete toughening [3,4], and structural material strengthening [4,5]. When used as anchoring bolts, the strength, durability, and other mechanical properties of BFRP should be carefully considered. In terms of FRP material strength and durability, researchers have conducted multiple relevant tests under naturally exposed environments and accelerated corrosion environments, respectively, aiming to explore FRP's resistance of acid, alkali, salt, water, and temperature [5–8], the results show that FRP is more easily corroded in alkali environment [9–11]. Compared to CFRP (Carbon Fiber Reinforced Polymer) and GFRP (Glass Fiber Reinforced Polymer), BFRP has stronger corrosion resistance [6,12,13], which can be intrinsically attributed to the existence of several oxides such as SiO₂ and Al₂O₃ [14]. In terms of the mechanical properties of anchorage, numerous laboratory and field tests and finite element modelling tests have been carried out [15–17]. It was found that for BFRP anchor bolts with a diameter of less than 20 mm, the failure mechanism of the

* Corresponding author. Chengdu Technological University Research Institute, Chengdu Technological University, Yibin, 644000, China.

E-mail address: weizhishuiyu@163.com (J. Chen).

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bolt-anchorage system is dominated by pull-up failure of the first interface (i.e. bolt-mortar interface), but the bolt itself is not destroyed [7,10,14,18,19]. By contrast, for anchor bolts with larger diameters (>20 mm), the failure mechanism is mainly manifested as shear failure in the second interface (i.e. mortar-soil interface), and the first interface is also destroyed [14,18–20]. This phenomenon occurs because the anchor diameter directly influences the stress concentration around the anchor hole. A smaller anchor diameter results in a thicker grouting layer, thereby amplifying the load transfer lag to the surrounding rock and soil. Under equivalent tensile loads, the anchoring system's first interface load-sharing is relatively higher. Conversely, for larger diameters, the load-sharing at the second interface becomes more significant. Based on the influence of factors such as mortar strength grade, bolt diameter, and effective anchorage length on the interface bonding strength, scholars have proposed several constitutive models, including Malver [21], CMR [22], BPE [23], and BPE modified models [24], the bond-slip model established for FRP reinforcement provides a theoretical foundation that is partially applicable to BFRP anchor bolt. However, extensive data are still required to validate the model's applicability and ensure the reliability of the theoretical formula [25, 26]. In terms of studies on the retaining and anchoring of rock and soil slopes, stability analysis and design methods are two research emphases. For instance, Cosenza [24] suggested to select FRP bars with a safety factor against pullout of 2.5 for pullout tests. By designing a BFRP bolt-anchorage system for soil slope, Guo [27] argued that a safety factor against pullout of 2.0 is more reasonable. Zhu [28] and Kang [8] conducted experiments on non-prestressing BFRP bars and proposed that the safety factor against pullout should be no less than 1.6 and 1.4 for permanent structure and temporary structure, respectively. Based on the Standard Codes of MOHURD [29], ACI [30] and ASTM [31], some researchers employed BFRP bars to replace regular reinforcement bars and analyzed the bearing characteristics and failure characteristics of BFRP concrete beams with different diameters [8,32,33]. In addition, many researchers have referred to the results of other types of FRP bars (e.g. bolt-mortar interface strength theory of GFRP; energy consumption theory of FRP bolt-mortar interface) for the design of BFRP bolt-anchorage system [16,17,25,34,35,36]. Experimental studies have shown that BFRP bolts used for slope anchoring are featured by low axial force attenuation rate [14,25], which confirm the feasibility of BFRP bars replacing regular reinforcement bars. However, due to the complexity of rock-soil masses and the variability in anchorage methods, current calculation approaches remain insufficient for accurately capturing the nonlinear characteristics of the BFRP bolt-anchorage system. Additionally, studies on the anchoring mechanism and performance of these interfaces often rely on laboratory pullout tests, which are significantly influenced by size effects and thus diverge considerably from practical engineering scenarios. The small diameters of anchors used in laboratory tests, coupled with the lack of consideration for in-situ stress conditions, limit the accuracy of these tests in reflecting the actual operational behavior of BFRP anchors. Therefore, it is essential to conduct field tests to effectively analyze the load transfer characteristics of the BFRP anchorage system and to establish a more appropriate model for the BFRP anchorage interface.

In this study, indoor pull-out tests of BFRP bolt-anchorage system were firstly conducted to investigate the bolt-mortar bonding strength of anchor bolts with different diameters and probe the failure mechanisms of the bolt-anchorage system. On this basis, a clay site was selected and excavated to form a clay slope. According to the equal strength replacement principle, an unsupported slope, a slope supported by BFRP bolt, and a slope supported by steel bolt were designed, respectively. Then, the horizontal displacement of surface and deep soil mass of slope, the difference of bolt axial force distribution during the “excavation - slope support - immersion-induced slope instability” process was analyzed, and the feasibility of using BFRP anchor bolt for slope was investigated. Finally, the failure mechanism of BFRP anchor bolt was discussed based on the deformation and failure characteristics of the

bolt-anchorage system.

2. Bonding performance and failure characteristics of the BFRP bolt-mortar interface

Previous studies have demonstrated that the failure mechanism of bolt-anchorage system and the slope stability are strongly dependent on the bonding performance of the bolt-mortar interface (the first interface, Fig. 1) [7,8,10,14,25]. This study includes two steps. Firstly, indoor pull-out tests were carried out to probe the tensile strength and failure characteristics of BFRP bars according the Code by MOHURD [29] and ACI [36]. Secondly, based on the methods proposed in the Code for design of concrete structures MOHURD [29], ACI [36,38], BFRP anchor bolts with 3 different diameters (8, 10, and 12 mm) and 3 grades of mortar (i.e. neat cement mortar; M20 mortar; M30 mortar) were selected for bonding strength test.

2.1. Tensile strength parameters of BFRP bolts

The BFRP bolts used in the test have diameters of 8, 10, and 12 mm, with three samples tested for each diameter. A total of 9 tensile strength tests were performed. The average densities of 2.080, 2.091, and 2.101 g/cm³, respectively. The average linear expansion coefficients are 9.86, 10.13, and $10.42 \times 10^{-6}/^{\circ}\text{C}$, respectively. Quartz sand is evenly distributed on the surface of the BFRP bolts, with no fiber protrusions and cracks, and the interface between fiber and epoxy is intact. The test was carried out using an NC (numerical control) hydraulic universal testing machine (UTM), on the basis of the tensile strength test standard of fiber material [29,36]. Due to the fact that the shear strength of BFRP bolts is weak and cannot be welded with steel tubes, epoxy glue was used to bond the steel tube (diameter (D) = 20 mm, thickness (d) = 3 mm, length (l) = 300 mm) with the two ends of a 1.2m long bolt (Fig. 2). The length of inserted bolt into the steel tube (l_1) is 15 cm, ensuring that the length of the middle bolt section is no less than 30 times of the diameter and no less than 500 mm. To enhance the friction between the inner wall of the steel tube and the epoxy glue, a conical convex was made on the steel tube wall every 5 cm using a hydraulic clamp. After 24 h, the prepared samples were installed on the testing machine, and a digital extensometer (Model No.-SCDY) was installed in the middle of the samples. Then, loading was performed at a rate of 2 mm/min for tensile strength testing until the samples were destroyed, load and strain data are collected synchronously using Static Data Acquisition System (TDS-300). The preparation and completion of the tensile strength test are shown in Fig. 2.

Fig. 3 shows the stress-strain curve of BFRP bolts obtained through tensile strength tests. The Y-axis refers to the average tensile stress obtained by dividing the load value of the cross-sectional area of BFRP bolts, while the X-axis represents strain that is directly recorded by TDS. According to Fig. 3, the tensile force corresponding to sample failure and the maximum tensile strength of the bars are obtained and listed in Table 1. It is noteworthy that the minimum tensile strength of the bolts is greater than 850 MPa, which is about twice that of ordinary steel bars. In

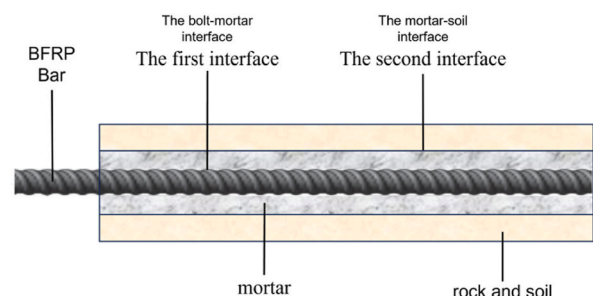


Fig. 1. Sketch map of bolt-anchorage system of BFRP.

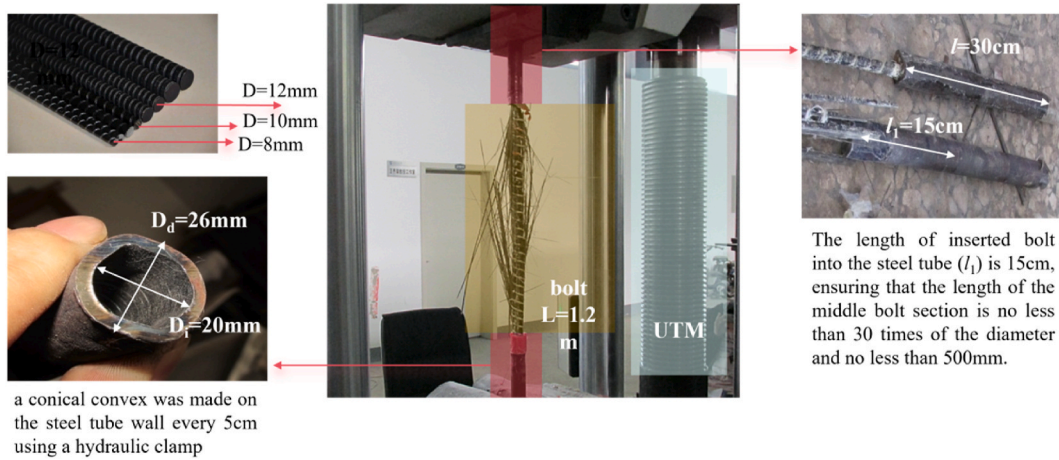


Fig. 2. UTM testing and illustration.

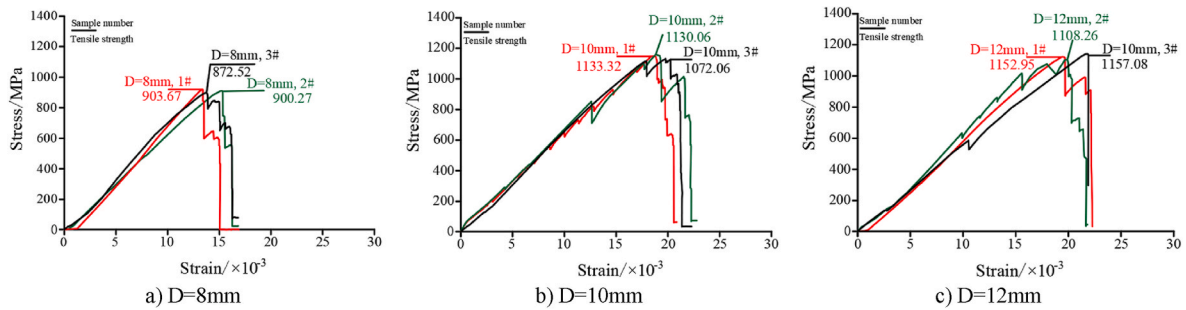


Fig. 3. The stress-strain curve of BFRP bolts.

Table 1

The tensile force corresponding to sample failure and the maximum tensile strength of the bars.

Test No.	Tensile force corresponding to sample failure /N	Average tensile force/N	Maximum tensile strength h/MPa	Average tensile strength/MPa
8-1	102,151	100726.5	903.67	891.1
8-2	101,767		900.27	
8-3	98,630		872.52	
10-1	88,966	87280.3	1133.32	1111.9
10-2	88,710		1130.06	
10-3	84,165		1072.16	
12-1	57,924	57,245	1152.95	1139.4
12-2	55,679		1108.26	
12-3	58,132		1157.08	

addition, the tensile elastic modulus of the bolts is greater than 45 GPa, which is about 1/4 of that of ordinary steel bars. These values meet the requirements of the tensile strength and material elastic modulus of BFRP bolts in the referenced specifications [29,36]. BFRP bar is mainly composed of basalt fibers and thermoset polymer. The tensile modulus of elasticity of a single basalt fiber can reach up to 110 GPa, which is much higher than that of polymer. Therefore, the tensile strength of BFRP mainly depends on the content of basalt fibers. The higher the content of basalt fibers, the greater the tensile strength of BFRP. However, excessively high fiber content results in low polymer content, making it difficult to extrude and solidify into shape. The content of basalt fibers per unit volume of BFRP with diameters of 8, 10, and 12 mm decreased, resulting in the strength of BFRP reinforcement not increasing linearly with diameter.

Theoretical stress-strain curve of BFRP bar can be simplified into a

trilinear curve (Fig. 4). During the test, when the strain reaches ε_1 , which is point An of the BFRP curve, it was found that glue peel-off occurred first on the surface of the BFRP bolts, followed by fiber pulling, that is, the epoxy was torn up and some fibers were broken, this is phase I; As the loading process continued, the sound of epoxy and fiber peel-off was heard. As the epoxy stripping ratio increases, basalt fibers will continue to bear all the load, the sound of fiber break became louder and more frequent after peak strength (point B), this is phase II; The load suddenly changes from point B to point C, and then enters the phase III, white spot-like cracks appeared on the surface of the BFRP bolts until the fibers suddenly broke, exhibiting an explosive pattern in the middle part of the BFRP bolts (Figs. 2 and 4).

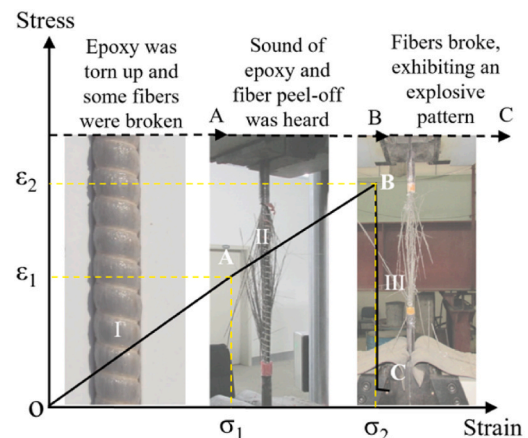


Fig. 4. Theoretical stress-strain curve of BFRP bar and its failure process.

2.2. Bonding performance between BFRP anchor bolt and mortar

For each pull out test samples (three different BFRP bar diameters: 8 mm, 10 mm, and 12 mm), three bolts were bonded to neat cement mortar, M20 mortar, and M30 mortar, respectively. A total of 27 pull-out tests were performed to evaluate the bonding performance, as shown in Fig. 5. According to the comparative test method for the bonding strength between steel bars and concrete in the Code for design of concrete structures [37,38], cylinder samples were prepared (Fig. 5). The length and diameter for the concrete cylinders are 250 mm and 110 mm, respectively, and the BFRP bolts are 550 mm long. In terms of the mortars: 1) The water-cement ratio of neat cement mortar is 0.5; 2) The ratio of cement, sand, and water of M20 mortar is 1:5:0.8; 3) The ratio of cement, sand, and water of M30 mortar is 1:3:1:0.6. The cement strength grades of the three types of mortars are all 42.5. After the prepared samples were cured for 28 days, steel tubes were installed as load-bearing ends (Fig. 2), and the pull-out test was carried out using a hollow jack (Fig. 5). During the test, the samples were loaded at a rate of 3 kN/min. The test was stopped when the bolts show obvious relative sliding or split-off relative to the samples. The comparison results of the bonding performance between BFRP bolts and different mortars are shown in Fig. 6.

As shown in Fig. 6, It is prominent that bolts with different diameters have different bonding strengths with mortar. Specifically, the bonding strength between BFRP bolt and neat cement mortar is approximately 3 MPa, which is about 40–50 % lower than the corresponding values of M20 and M30 mortars (5–6 MPa). This is due to the differences in material composition and viscosity of mortars with varying grade. Compared with the designed value recommended in the specification GB50330 [39], the bonding strength between BFRP bolts and mortars is larger under the same grade. For example, the bonding strength between BFRP bolts and neat cement mortar is higher than the minimum value (2.1 MPa) recommended in the specification GB50330 [39], and the bonding strength between BFRP bolts and M30 mortar is greater than 5 MPa (about twice that of steel anchor bolt). Overall, the anchorage force increases with the increase of bolt diameter and cement mortar strength. This is because when the strength of cement mortar increases, the internal crystal structure and cementation material become denser and harder, which improves the compressive strength, tensile strength, and toughness of mortar.

During the bonding strength test, obvious relative slip failure occurred between the free end of BFRP bolts and concrete cylinder (solidified mortar) (Fig. 7), accompanied by the fiber break of the bolt outer surface. During the failure process, as the loading increased, the end of the anchorage body initially broke, generating small cracks. Then, slight sound of anchor bolt break was heard. With the continuous increase of loading, the sound of fiber break became louder and more

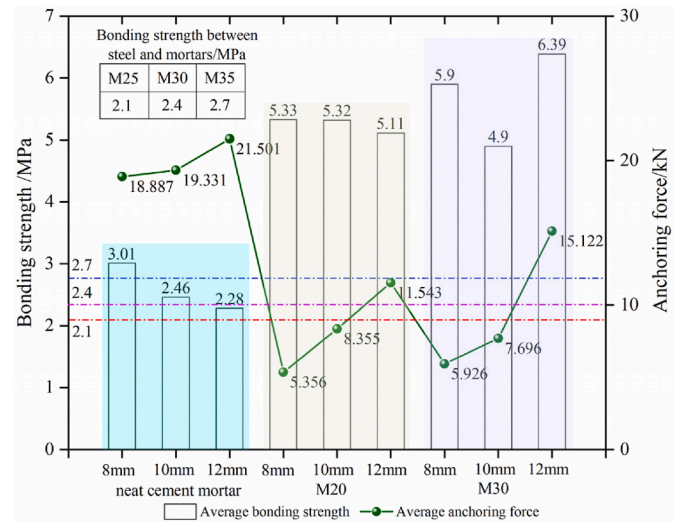


Fig. 6. The comparison results of the bonding performance between BFRP bolts and different mortars.

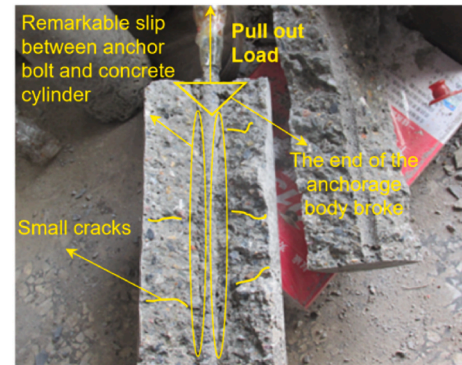


Fig. 7. Relative slip failure characteristics.

frequent. Meanwhile, cracks began to appear on the surface of the anchor bolt, indicating that the anchor bolt gradually lost its bearing capacity. Finally, remarkable slip was observed between the anchor bolt and concrete cylinder.

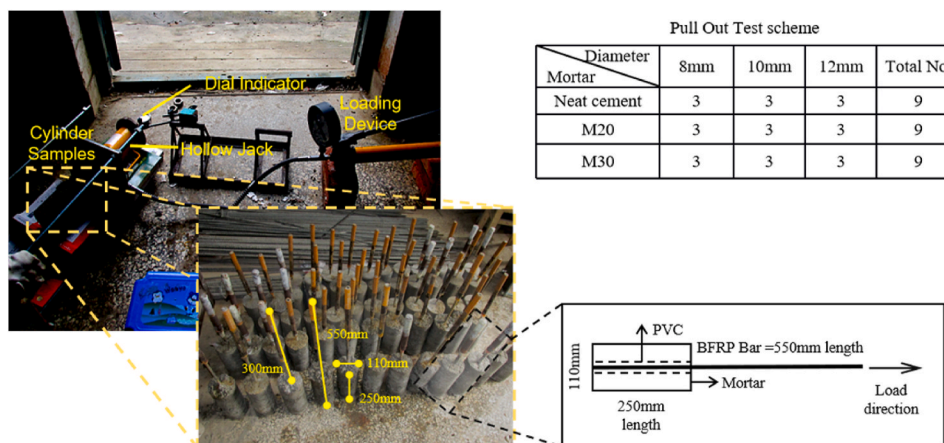


Fig. 5. Bonding performance test—pull out test of cylinder samples.

3. In-situ test on the anchorage performance of BFRP bolt-anchorage system

3.1. Test scheme

3.1.1. Slope excavation

Based on the bonding performance results (Section 2), in-situ tests were carried out to probe the anchorage performance of BFRP bolts. The test site is the third-level terrace of the Min-jiang River system in the Chengdu Plain. The soil layers in the test site are as follows: 1) Quaternary Holocene artificial fill (Q_4^{ml}): ①3–4m thick recent fill (Q_4^{ml-1}): mainly composed of construction wastes intercalated with pebbles; characterized by soft texture, poor consolidation, and low strength; ② 0.5–1m thick old fill (Q_4^{ml-2}): mainly composed of clay soil intercalated with silt soil; characterized by low strength and high compressibility; 2) Quaternary Middle and Lower Pleistocene aqueoglacial sedimentary layer (Q_{2-1}^{gl}): clay layer with weak expansion potential, containing iron-manganese oxide nodules and a small portion of calcareous nodules. Net-shaped fissures filled with thin gray clay layers are developed. It is notable that no groundwater was found within a depth of 20m underground in this region. According to the engineering characteristics of this site, the length, width, and depth of the top of the tested slope were determined to be 22m, 22m, 9m, respectively. The site was excavated in three levels. The third-level slope is 3.2m high with a slope rate of 1:1. The second-level slope is 2.6m high and the slope rate is 1:0.5, and a 1m wide platform is set at the foot of this slope. The first-level slope is 3.2m high and the slope rate is 1:0.5. The maps of the test slope are shown in Fig. 8. According to the risk levels corresponding to the damaged anchor bolts, the safety level is determined to be III, and the minimum safety factor of temporary anchor bolts is suggested to be 1.4 [28,40,41].

3.1.2. Slope retaining

The third-level slope is composed of recent fills, which have poor engineering properties due to the existence of considerable construction wastes. Side slope collapses may occur in the bolt-anchorage support process. Thus, soil nailing was employed for slope support in this study according to specification JGJ120 [42].

When designing the anchor bolts for the first- and second-level slopes, the soil of the third-level slope should be considered in a steady state after nailing (safety factor of slope stability k is greater than 1.2). Thus, non-prestressing steel bolt support scheme was selected according to Zhu et al. [28] and specification CECS 22 [40] for the left slope in Fig. 8. The steel bolt was designed based on the material tensile strength within its service life. The BFRP bolts for the first- and second-level slopes were designed (the right slope in Fig. 8) based on the equal strength replacement principle that describes the relationship between steel and BFRP anchor bolts (Eq. (1)) [8,33].

$$f_{sd\text{steel}}S_{\text{steel}} = f_{sd\text{BFRP}}S_{\text{BFRP}} \quad (1)$$

where $f_{sd\text{steel}}$ is the tensile strength of steel bars (N/mm^2). S_{steel} is the cross-section area of steel bars (mm^2). $f_{sd\text{BFRP}}$ is the tensile strength of the BFRP (N/mm^2). S_{BFRP} is the cross-section area of the BFRP (mm^2).

The tensile strength of FRP bars could be calculated using Eq (2), which is based on specification ACI 440. 1R-2015 [30]. Eq (3) is used to calculate the anchorage length [43], and the standard values of bonding strength can be found in section 2.1.

$$f_{sd\text{BFRP}} = f_{\text{avg}} \cdot 3\sigma \quad (2)$$

$$L_a \geq \frac{K \cdot N_t}{\pi \cdot D \cdot f_{mg} \cdot \phi} \text{ or } L_a \geq \frac{K \cdot N_t}{\pi \cdot \pi \cdot d \cdot \xi \cdot f_{ms} \cdot \phi} \text{ (the maximum value)} \quad (3)$$

where f_{avg} is the average value of the limit tensile strength of the anchor

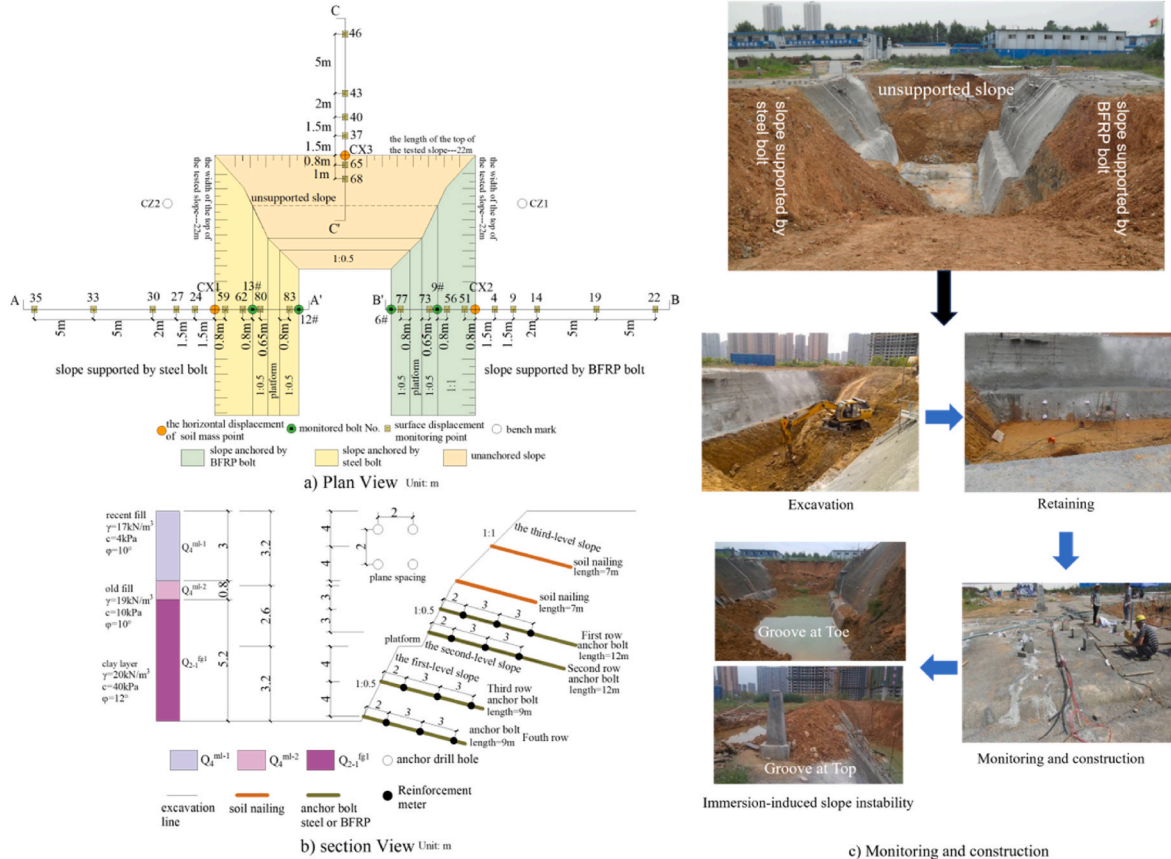


Fig. 8. The maps of the test slope (unit: m).

bolt (N/mm^2). σ is the standard deviation of the limit tensile strength (N/mm^2). L_a is the length of the bolt-anchorage system (m). K is the safety factor against pullout. f_{mg} is the standard value of the bonding strength between anchorage mortar and surrounding strata (N/mm^2). f_{mb} is the standard value of the bonding strength between anchorage mortar and anchor bolt (N/mm^2). D is the borehole diameter of the bolt-anchorage system (mm). d is the bolt diameter (mm). ξ is the decreasing coefficient of the interface bonding strength, with a range of 0.6–0.85. φ is the influencing factor of the anchorage length on the bonding strength. n is the number of BFRP bolts. N_t is axial tension of bolt (kN).

After calculation, the total sliding force was 445.393 kN/m and the total anti-sliding force was 404.272 kN/m, and thus the required reinforcement force for each linear meter of slope was 160 kN. Therefore, the designed value for the anchoring force of the first and second bolt rows was 70 kN (they are in the second-level slope), and for the third and fourth bolt rows was 60 kN (they are in the first-level slope). Detailed design parameters are listed in Table 2, and the schematic diagram of slope support are shown in Fig. 8b.

3.2. Water-immersion destructive test

The experimental design was initially intended to monitor slope deformation and failure during natural rainfall in the rainy season. However, constrained by time limitations, a direct water immersion method was employed to induce deformation and failure within a shorter timeframe. The experimental design is as follows: After routine monitoring period, the destructive test of the slope was conducted by excavating the slope toe and slope top, and then immersing. Specifically, grooves with a depth of 1.5m and a width of 1m were excavated on the slope toe and slope top. Then, the grooves were immersed in water for 1–2 weeks (Fig. 8c). During the slope failure process, slope deformation and the force state of anchor bolts were synchronously monitored.

3.3. Monitoring scheme

The monitoring contents include the ground horizontal displacement of the slope, the deep horizontal displacement and the axial force of anchor bolts, which are automatically measured by total stations, inclinometers and steel bar meters. The layout of relevant monitoring points is shown in Fig. 8a, the actual photos of the monitoring system are shown in Fig. 9.

- 1) The ground horizontal displacement monitoring system consists of two base points, two total stations, and several surface monitoring points. The base points are located outside the slope, while stations CZ1 and CZ2 are located at the top of the two supported slopes, close to the unsupported slope. For each test slope, an observational profile was set, which named section A-A' for slope supported by steel bolt, section B-B' for slope supported by BFRP bolt, section C-C' for unsupported slope, respectively. The horizontal displacement monitoring points on the slope top were buried after the excavation and leveling of the third-level slope, the deformation monitoring points was arranged and buried on site after excavation for each level of slope. According to the stability of the slope after excavation, 9 monitoring points are buried in section A-A' and B-B', and 6

monitoring points are buried in section C-C'. The location of displacement monitoring points is shown in Fig. 8a.

- 2) Three inclinometers (CX1 for the slope supported by steel bolt; CX2 for the slope supported by BFRP bolt; CX3 for unsupported slope) were employed to measure the horizontal displacement of deep soil mass of the slopes. Both inclinometers are 14m long, and the depth interval between measuring points is 1m. The location of monitoring sites is shown in Fig. 8a.
- 3) The axial force of anchor bolts was monitored by steel bar meters. There are 4 bolts installed with steel bar meters: two of them are for the first-level slope, while the other two are for the second-level slope. The steel bar meters for steel bolts were numbered 12# and 13#, while those for BFRP bolts were numbered 6# and 9#. For each individual bolt, three steel bar meters were installed inside, and the positions are 2m, 5m, and 8m away from the anchor head (as seen in Fig. 8b). It should be noted that the steel anchor bolts were welded to the steel bar meters. However, BFRP anchor bolts cannot be welded to steel tubes (steel bar meters) due to its weak shear resistance, and thus we used epoxy glue to bond them in this study. For detailed operational procedures, please refer to Section 2.1.

The construction method utilized in the test is layered excavation, and the process can be summarized as follows: excavation, retaining (hole drilling, anchorage, grouting), net hanging, guniting, and re-excavation (as seen in Fig. 8c). After each level of slope excavation was completed, slope deformation and the axial force of anchor bolts were immediately monitored. The stable monitoring period of each level of slope is 7–10 days. The stability standard is based on specification GB50330 [39], and the monitoring frequency is based on specification JGJ120 [42], which is 1 time/day for data collection. After the end of the routine monitoring period, if there was no slope instability occurred, slope failure was induced using the method described in Section 3.2.

3.4. Test results and analysis

3.4.1. The ground displacement - loading-time relationship

After each level of slope excavation was completed, the step of bolt support was carried out. The total monitoring period that includes slope excavation, slope support, groove immersion is about 70 days. The focus of this study is the difference in the retaining capacity of BFRP and steel bolts. The starting point of the monitoring data is dependent on the excavation time of the second-level slope (May 7th, 2023). The surface horizontal displacement against time curve is shown in Fig. 10, the horizontal axis represents the days after excavation of the secondary slope.

- 1) The surface deformation characteristics of the supported slope is shown in Fig. 10a. Since excavation, the slope deformation has increased at a large rate. After the excavation of the first-level slope, the maximum horizontal displacement of the surface reached nearly 10 mm. On the 10–14th day after the excavation of the first-level slope, the horizontal displacement of the surface increases to 20 mm linearly, then it instantly became unstable and destroyed (Fig. 11a). The failure mechanism was dominated by slip failure, with the top of the slope moving 5.2m downward.

Table 2
Slope retaining scheme.

Retaining type	d/mm	L_a /m	S/(m × m)	$\theta_i/^\circ$	D/mm	K	Mortar
BFRP	14	12; 12; 9; 9	1.5 × 1.5	15	100	1.4	M30 (grouting of total length)
Steel (HRB335)	25						
Soil nailing	Inner-d = 26 Out-d = 30	7; 7	1.5 × 1.5	15	/		

12; 12; 9; 9 refers to the length of the first, second, third and fourth row of anchor bolts being 12m, 12m, 9m, and 9m, respectively. d is the bolt diameter, D is the borehole diameter of the bolt-anchorage system, L_a is the length of the bolt-anchorage system, S is plane spacing of the bolts, θ_i is installation angle.

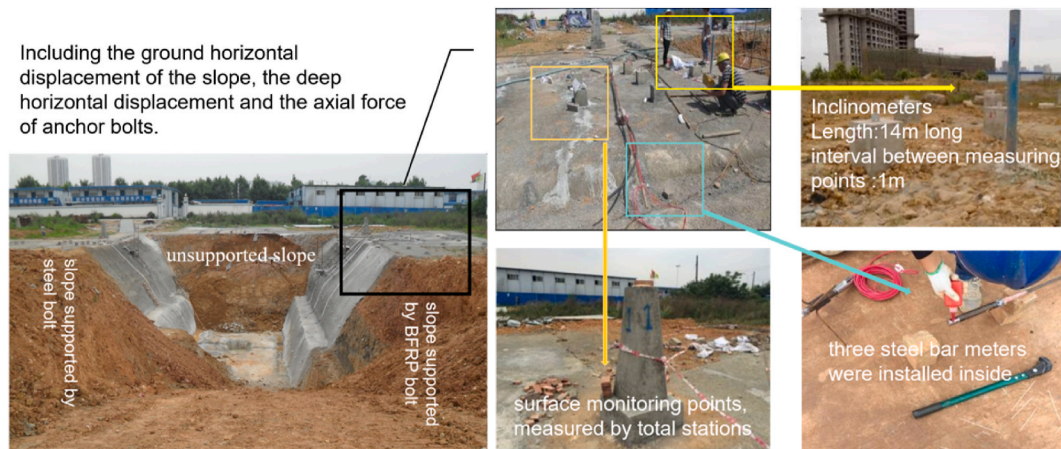


Fig. 9. Monitoring and construction.

- 2) The surface deformation of the slope supported by BFRP anchor bolt is shown in Fig. 10b. After the excavation of the first-level slope, the maximum horizontal displacement of the monitoring section occurred at the top of slope, reaching about 10 mm; The deformation rate of the slope decreased and gradually became stable with slope anchoring. Before destructive test of slope by immersing (On the 47th day after the excavation of the first-level slope), the maximum horizontal displacement at the rear edge of the slope top was only 12 mm (a deformation rate of 0.05 mm/d); When the toe of slope was excavated and immersed in water, and the slope deformation increased rapidly to 24 mm immediately. 5 days later, the horizontal displacement of the slope increased to 37 mm, corresponding to a deformation rate of 2.6 mm/d. On the 4th day after the top of slope was grooved and immersed in water, and the maximum regional displacement reached 40 mm, and 14 days later the maximum regional displacement reached 80 mm which is approaching the allowable displacement of the slope, corresponding to a deformation rate of 2.38 mm/d. Then, several tensile cracks appeared in the range that is 0.5–1.5m away from the slope top, but no obvious slip failure was observed (Fig. 11b).
- 3) The surface deformation of the slope supported by steel anchor bolt is shown in Fig. 10c. After the excavation of the first-level slope, the maximum horizontal displacement of the monitoring section occurred at the top of the slope, reaching approximately 12 mm. After the support of steel anchor bolts, the deformation rate of the slope decreased and gradually became stable. Before destructive test of slope by immersing (On the 47th day after the excavation of the first-level slope), the maximum horizontal displacement at the rear edge of the slope top was 18 mm (a deformation rate of 0.15 mm/d). When the toe of slope was excavated and immersed in water, and the slope deformation increased rapidly to 28 mm immediately. Before the top of slope was grooved and immersed in water, the horizontal displacement of the slope has been increased to 59 mm already, corresponding to a deformation rate of 3.87 mm/d. On the 8th day after slope top grooved, and the maximum displacement reached 80 mm which is approaching the allowable displacement of the slope, corresponding to a deformation rate of 2.62 mm/d. Then, a penetrated tensile crack appeared in the range that is 0.5m away from the slope top, (Fig. 11c).

3.4.2. The deep horizontal displacement - loading-time relationship

The horizontal displacement curve of deep soil was shown in Fig. 11. Three inclinometers on the slope top were buried before the excavation and leveling of the third-level slope.

For the slope supported by BFRP bolts and steel bolts, it can be concluded the laws of deep horizontal displacement with the depth is the

same that during the excavation of the two slopes with two types anchoring, the maximum soil horizontal displacement approaching the surface, the deformation in the range within a depth range of 7–8m from surface was prominent, and increased with the increase of excavation depth. The primary slip surface is 6m underground, and the secondary slip surface is 8m underground. Overall, the deformation and failure mechanisms of these two types of supported slopes are similar. However, the displacement of steel-supported slope was larger than that of BFRP-supported slope, and multiple penetrated cracks appeared at the top, indicating that the critical failure state of slope was reached.

3.4.3. The loading-time-axial force relationship of anchor bolt

The axial force against time curves of the BFRP bolt region (6#) and the steel bolt region (12#) of the first-level slope are shown in Fig. 12a and b, respectively, while those of the BFRP bolt region (9#) and the steel bolt region (13#) of the second-level slope are shown in Fig. 12c and d, respectively.

As shown in Fig. 12, the axial force change trends of BFRP and steel anchor bolts are basically the same. After anchor bolt grouting is completed, the axial force gradually increases and then becomes stable. The maximum axial forces of the BFRP and steel bolts appeared at a distance of 2m and 5m from the slope surface, respectively, corresponding to the middle part of the axial force monitoring section. The stability value of axial forces at the distance 5m of anchor bolts in the second-level slope are 17.8 kN and 15.3 kN for BFRP bolts and steel bolts, respectively, and those in the first-level slope are 5.2 kN and 23.5 kN, which are all far smaller than the designed values.

When the slopes entered the destructive test stage, the force on both BFRP and steel anchor bolts increased rapidly. The axial force in the locations of the steel anchor bolt exceeded the designed value on the both 1st - and 2nd level slope, and that of BFRP anchor bolt only reached more than 95 % of the designed value. Specifically, after groove immersion at the slope toe, the maximum tensile force along the steel bolts and BFRP bolts of the first-level slope reached 43.5 kN and 14.7 kN, respectively, at 5m from the slope surface. By contrast, after groove immersion at the slope top, the maximum tensile force along the BFRP bolt was close to 60 kN, which is 97 % of the designed value, while the maximum tensile force along the steel bolt was 87 kN, which is 45 % larger than the designed value (60 kN). For the second-level slope, the maximum tensile force along the steel bolt reached 36.5 kN at 5m from the slope surface after groove immersion at the toe of slope, that is 38.9 kN for BFRP bolt. As of groove immersion at the top of slope, the maximum tensile force along the BFRP bolt was close to 68 kN, which is 97 % of the designed value (70 kN), the maximum tensile force along the steel bolt was 97 kN, which is 40 % larger than the designed value. These comparison results, when combined with the surface deformation and

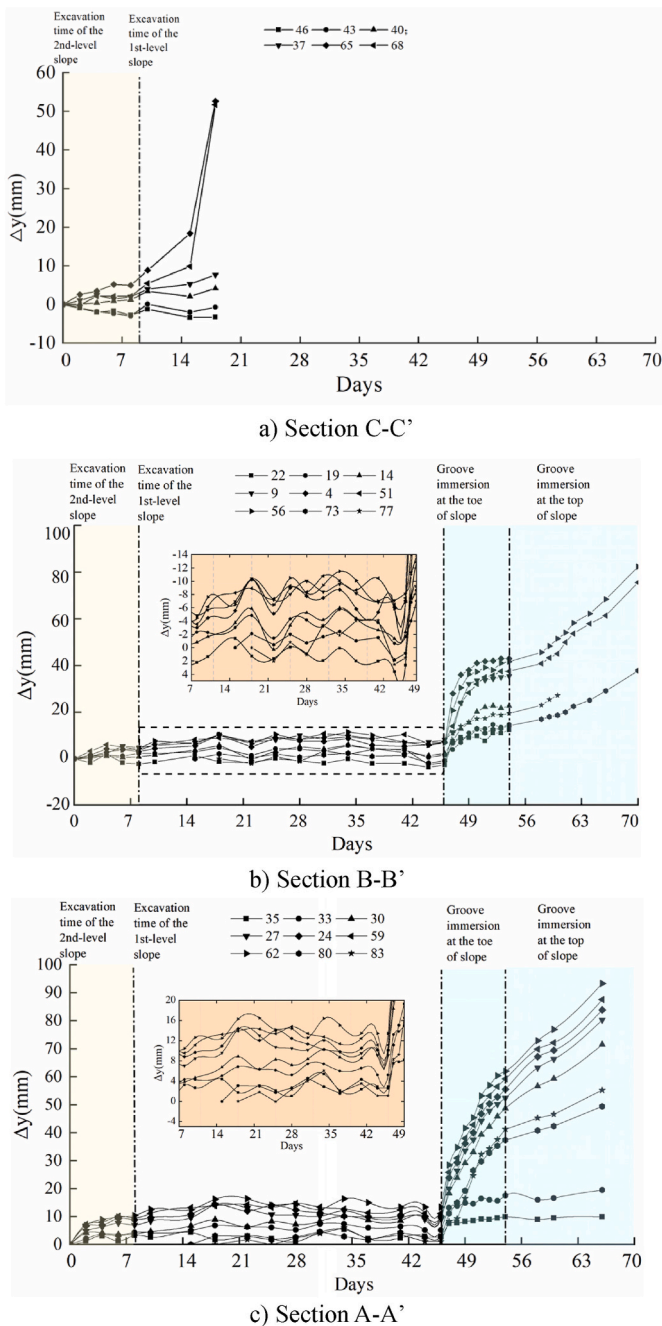


Fig. 10. The ground displacement - loading-time curve.

failure characteristics of the slope described in previous Sections, suggest that the slope supported by steel anchor bolt was destroyed before the slope supported by BFRP anchor bolt.

3.4.4. Difference between slopes retained by different bolts

By comparing the monitoring results of the axial force and horizontal displacement of slopes supported by the two types of anchor bolts, it is obvious that both slope displacement and the axial force of the BFRP anchor bolts are smaller than those of the steel-supported slope. The slope deformation and failure can be divided into two stages (Fig. 13), which are the stable creep stage and the accelerated creep stage, respectively. The slope anchor performance of the two types of bolts is different:

During the stable creep stage, the axial force of BFRP anchor bolts and the corresponding slope displacement are similar to those associated

with steel anchor bolts. In this situation, BFRP is used to replace steel to ensure that the two have similar capabilities in resisting tensile force [8, 14,18]. Additionally, the axial force is only 1/3–1/4 of the designed value of anchorage force, and the deformation is only 12 mm, which meet the requirements of specification JGJ 120 [42].

During the accelerated creep stage, the displacement and axial force of BFRP anchor bolts is less than that of the steel anchor bolts. Although the displacement and tensile force of BFRP anchor bolts continued to increase induced by slope failure process, the maximum horizontal displacement at the rear edge of the slope top was increasing at a small rate, which is approximately 2/3 that of steel bolt, and the bolt axial force value did not exceed the designed value of anchorage force until the end of the test. By contrast, the maximum tensile force along the steel bolt is 40 % larger than the designed value, the maximum displacement reached 80 mm which is approaching the allowable displacement of the slope, corresponding to a deformation rate of 2.62 mm/d.

Consequently, the slope supported by steel bolts lost its bearing capacity and experienced overall instability that has similar characteristics with those of the unsupported slope. Therefore, it can be concluded that BFRP anchor bolts have a better performance in controlling slope deformation, and the long-term support effect and support stability of BFRP bolts on soil slopes are better than those of steel anchor bolts.

4. Discussion

4.1. Failure mechanism of BFRP bolt-anchorage system

Failure process of the bolt-anchorage system consists of two stages, there are the bonding failure between BFRP bolt and anchorage mortar, then the adhesive failure between the bolt fibers and epoxy matrix.

When subjected to external loading, all the force types of chemical adhesion, interlock and friction play a role in bearing the load. At the initial stage of loading, the shear strength between bolt and mortar is less than the bonding strength between them, and the chemical adhesive force and friction force play the key roles. As the loading increases and the friction and chemical adhesive forces are exceeded, the mechanical bite force begins to bear most of the load. As the loading further increases, the bolt surface threads are thoroughly abraded from shallow to deep, and cracks inside the solidified mortar expand outward. When the loading force exceeds the bolt-mortar bonding strength, debonding and slip failure can be observed between the BFRP anchor bolt and the surrounding mortar [20,21]. Particularly, the bolt-anchorage system broke explosively outward (Fig. 14 left) in the test conducted by Bai et al. [20], which can be attributed to the huge difference between the bonding and confine capacity inside and outside the bolt-anchorage system. However, such a phenomenon was not clearly observed in this study and may need more tests with higher precision.

By analyzing the shear failure mechanism of BFRP anchor bolt, it was found that the bolt fibers and epoxy matrix jointly bear the axial pull-out load. The failure is manifested as fiber pull-apart and tensile break of the middle bolt section, with an explosive shape (Fig. 4). Assuming that the deformation of the fibers and epoxy matrix is uniform under the axial pull-out loading, when the local stress of outer fibers reaches the failure stress threshold, these fibers will break, leading to the stress redistribution. At the same time, the stress will be transferred to the inner fibers, resulting in successive break of these fibers. Due to the random breaking location of fibers, the overall bearing capacity of the anchor bolt will not be seriously affected even if multiple fibers break. In addition, the regions where fiber breaks are concentrated are often subjected to large stress, and the remaining number of fibers are not able to bear the pull-out load, thus leading to the pull-out failure of the bolt body or the bolt-mortar interface (Fig. 14, right).

H.Wei et al. [25] also explained the bonding performance and load transfer characteristics of BFRP anchors, the pull-out load is mainly supported by the mortar holding force and mechanical interlocking at

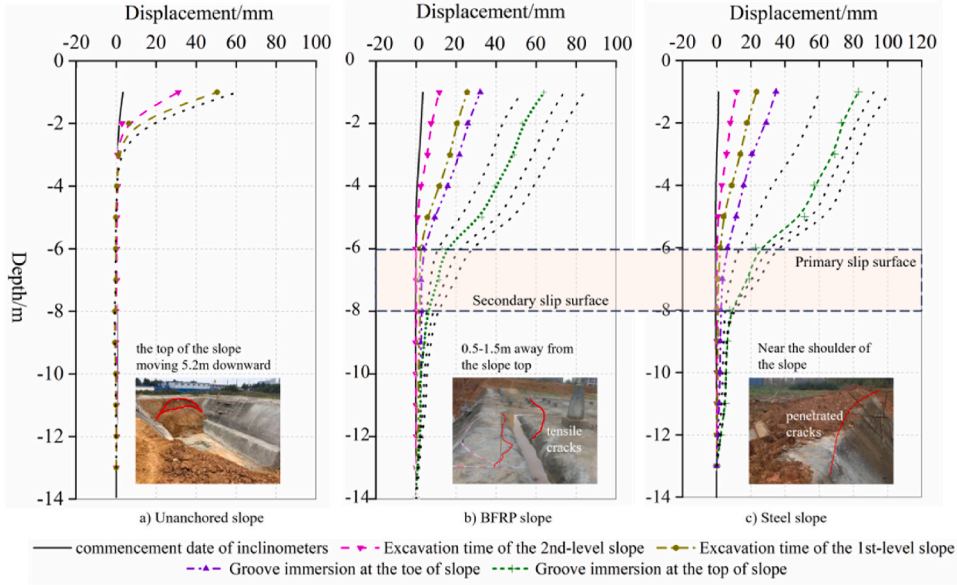


Fig. 11. The horizontal displacement curve of deep soil.

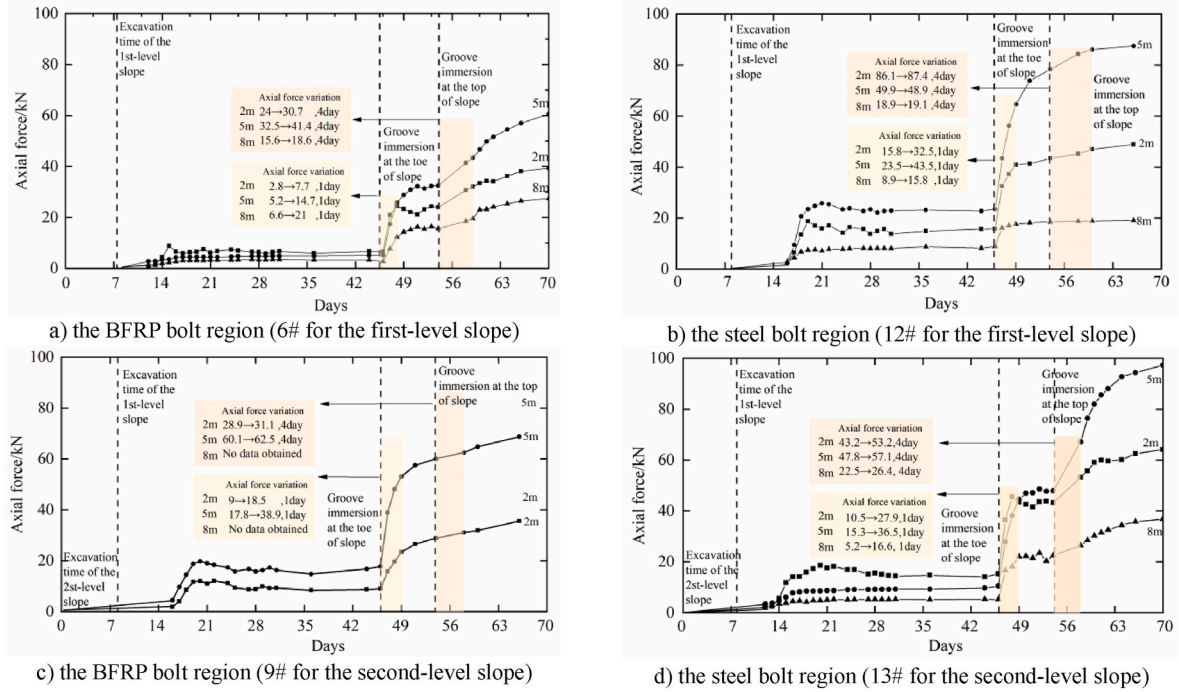


Fig. 12. The axial force against time curves.

the first anchorage interface, which gradually transfers the load to the grouting body initially. Subsequently, the cohesion and friction force at the second interface facilitate the transfer of the pull-out load to the rock-soil mass. Consequently, both the first and second anchorage interfaces are potential weak areas that may experience slip failure due to insufficient shear strength. This destructive mechanism is analogous to the phenomenon observed during our experiments.

4.2. Safety factor of tensile strength of BFRP bolt

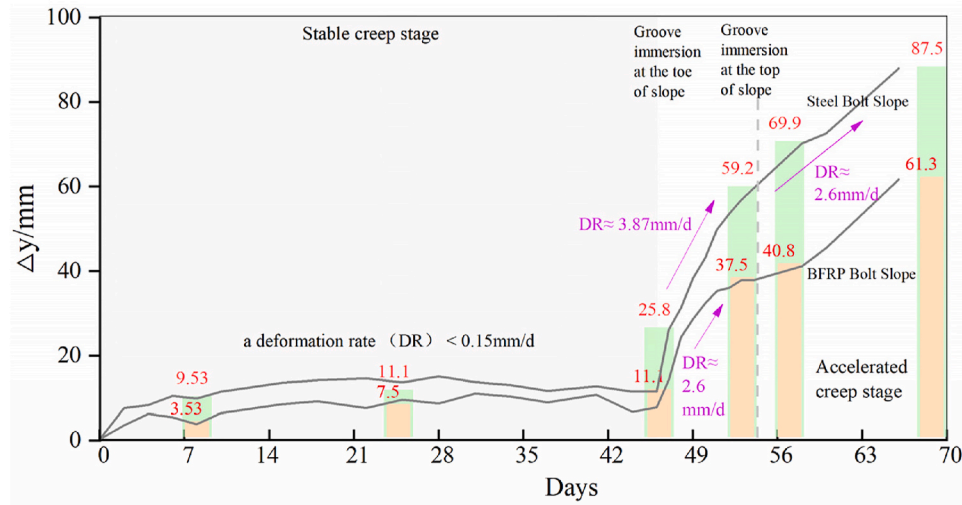
According to specifications [40,43], the safety factor (K) of anchor bolt can be obtained through Eq (4) and Eq (5).

$$K = \frac{T_u}{T} \quad (4)$$

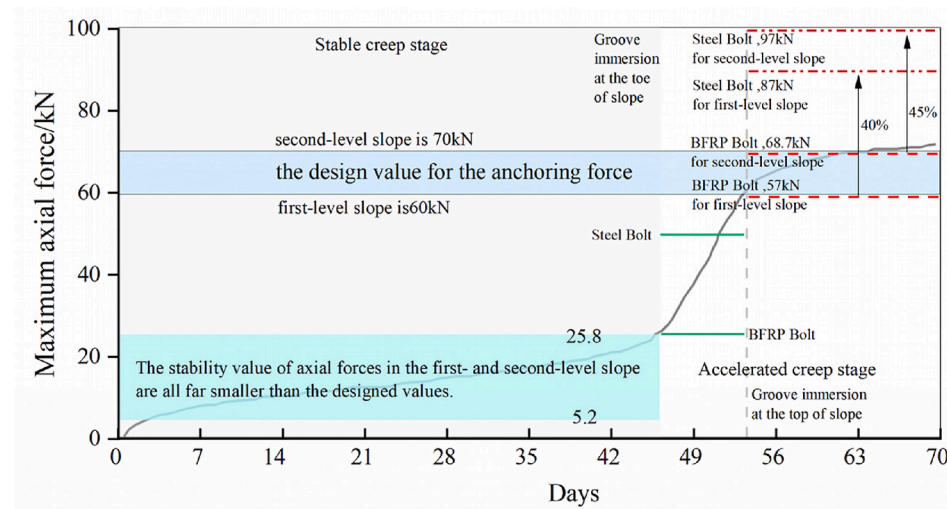
$$T_u = 0.95P_{\max} \quad (5)$$

where, T_u is the corresponding pulling resistance value (kN), T is the design tensile force of bolt (kN), $P_{\max}$ is the failure load of steel bolts.

It was suggested that when conducting BFRP anchor bolt design, both the safety factor based on bolt-mortar bonding strength and the safety factor associated with soil-mortar bonding strength should be considered [7,10,14,44]. In addition, the specification [43] suggested that the safety factor of tensile strength of bolt calculated based on the bonding strength of bolt-anchorage system is larger than that calculated



a) Displacement



b) Axial force

Fig. 13. Comparison of experimental results between two types bolts.

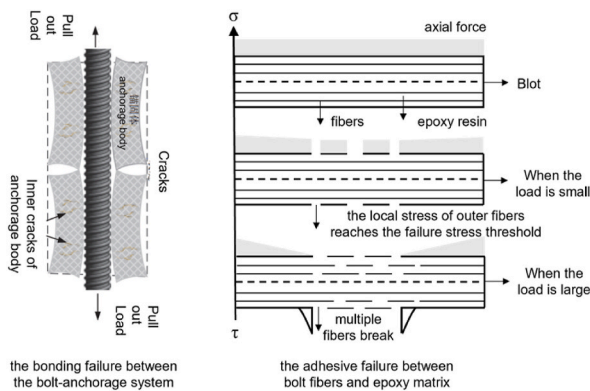


Fig. 14. Failure mechanism of BFRP bolt-anchorage system (Bai et al., 2024 [20]).

based on the tensile strength of anchor bolt. For the safety factor of tensile strength of BFRP bolts, engineers are recommended to refer to the specifications for steel anchor bolts. Thus, the safety factors under

different working conditions were back-calculated according to the in-situ test results, aiming to discuss the rationality of the recommended safety factor value.

- ① The instability and failure of common supported slopes are caused by the shear slip between the bolt-anchorage system and the surrounding soil. However, no obvious damage was found in the slope region supported by BFRP anchor bolts, indicating that the corresponding pulling resistance value (T_u) was greater than 68 kN. Limited by the equal strength replacement principle, the failure load of BFRP bolts (P_{max}) was less than or equal to 98 kN (the failure load of steel bolts). Taking 98 kN as the upper limit value, we considered the bolt-mortar and mortar-soil bonding strengths, and the bolt pull-out safety factor was back-calculated to be 1.55 ($P_{max} = 98$ kN; $T_u = 93$ kN; $T = 70$ kN), which is greater than the designed minimum value of 1.4.
- ② The standard value of the tensile strength of the 14 mm-diameter BFRP anchor bolt is 730 MPa, and the limit tensile strength is 917 MPa. At the end of the test, the tensile stress of the bolt was only 422 MPa. According to Eq (4), the safety factor of BFRP anchor bolt is 1.26, which is obviously large. This is mainly because that

the calculation was based on the reliable strength, that is used as the standard value of FRP tensile strength. However, the properties of FRP bars vary greatly. Zhu [28] conducted experiments on non-prestressed BFRP bars and recommended that the safety factor against pullout should be at least 1.6 for permanent structures and 1.4 for temporary structures. Studies have shown that this method overestimates the standard value of the BFRP tensile strength [8,45,46]. Therefore, the minimum safety factor can be adjusted to 1.26.

5. Conclusions

- 1) Under the condition of same bolt diameter and mortar strength grade, both the tensile strength of BFRP bolts and the bonding strength of the first interface are twice those of steel bolts. The bonding strength tests show obvious slip failure in the first interface of the BFRP bolt-anchorage system, and the failure mechanisms mainly include shear failure in the bolt-mortar interface and even pull-out failure of the bolt body, which are essentially caused by the cumulative damage of polymers.
- 2) The BFRP bolt-anchorage system designed based on the equal strength replacement principle of the bonding strength parameters can ensure the safety of slope engineering. During the stable creep stage, the bolt axial force and slope deformation characteristics of BFRP anchor bolts are consistent with those of steel anchor bolts, exhibiting excellent anchoring performance. From the accelerated creep stage to slope failure, both the bolt axial force and slope deformation continuously increase. Compared with the slope supported by steel bolts, the slope supported by BFRP bolts are always in a stable stage in the entire life-cycle of the slope excavation process, and the axial force of steel anchor bolts exceeded the designed value by 40 %. Furthermore, the axial force of BFRP bolt is less than the designed value and is only 2/3 of that of steel bolt, indicative of a better slope support capacity.
- 3) Based on the results of pull-out tests on BFRP bolt cylinders and in-situ tests on BFRP anchorage performance, as well as the structural reliability index value, the minimum safety factor of tensile strength of BFRP bolt of temporary slope is recommended to be 1.26, which is less than the designed safety factor value (1.4).

CRedit authorship contribution statement

Jibin Chen: Writing – review & editing, Writing – original draft. **Xiaobo Lv:** Writing – review & editing. **Jingwen Kang:** Validation, Investigation. **Guanglei Liu:** Supervision. **Yongxiang Zhou:** Supervision.

Data availability statement

All data, models, and code generated or used during the study appear in the submitted article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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